

TOTAL COMMERCIAL FLIGHTS

7,080,000

US Bureau of Transportation Stats

ON-TIME PERFORMANCE (OTP)

78.41%

5,463,700 Flights On Schedule

SYSTEM DELAY RATE

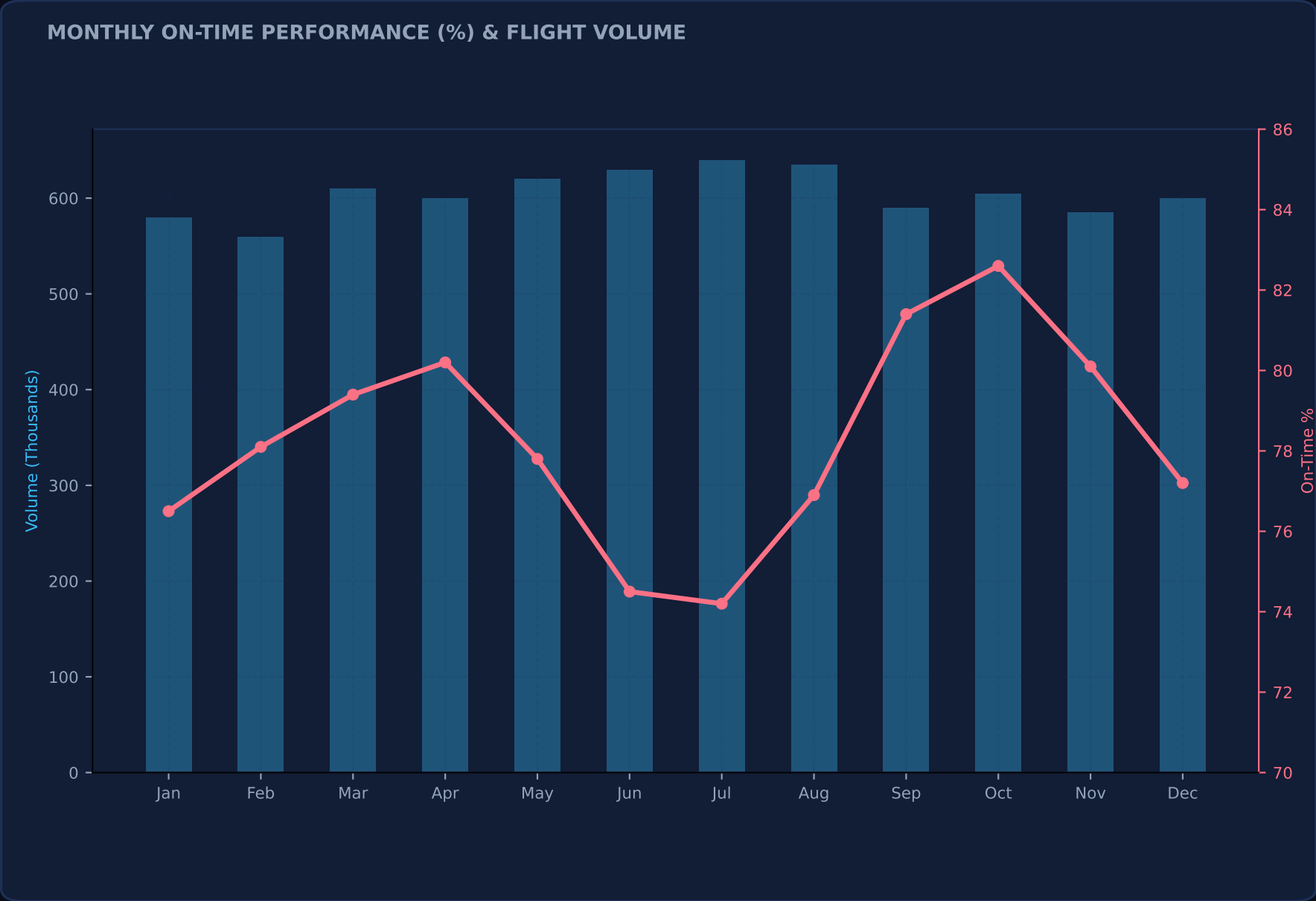
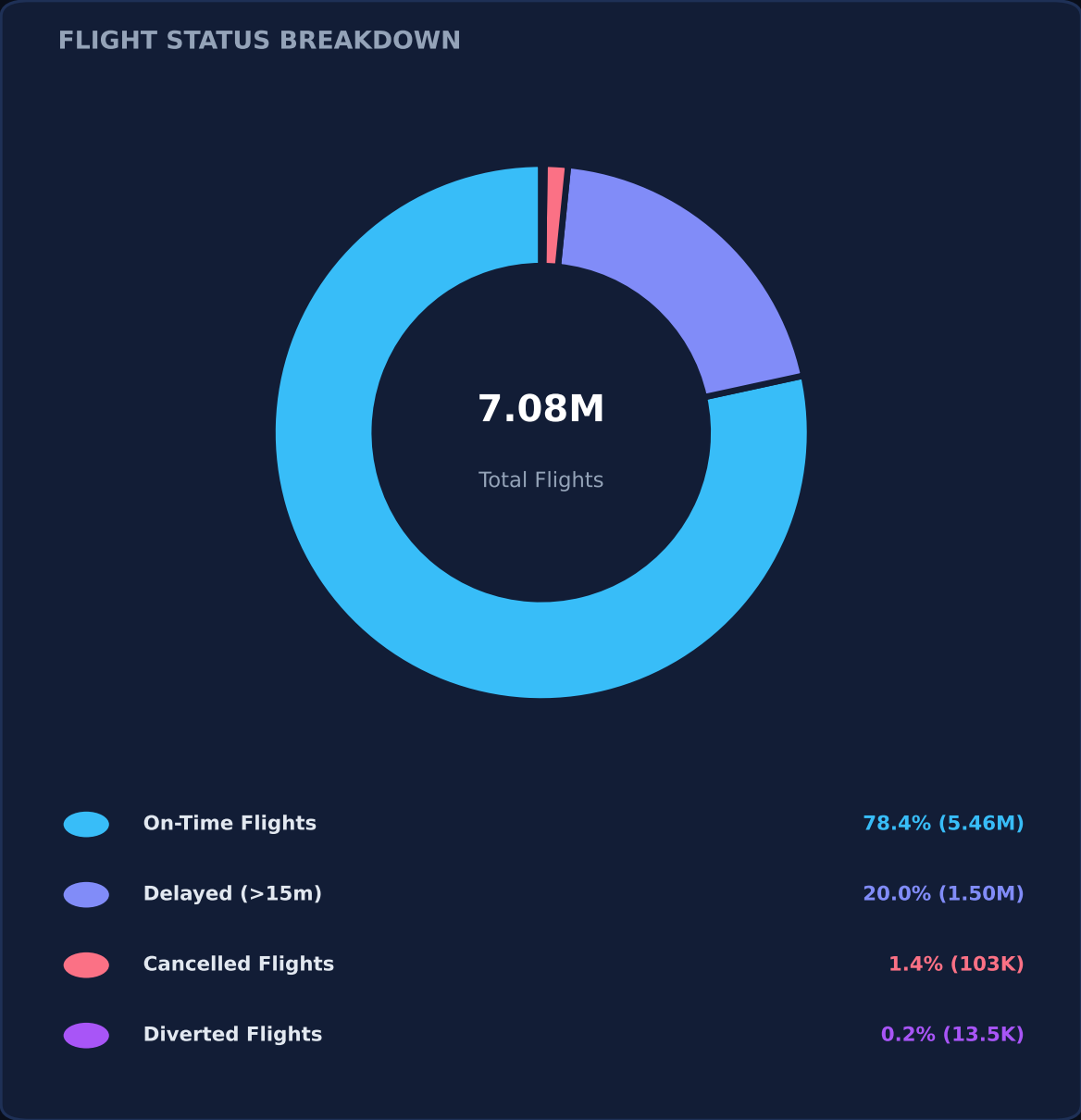
21.59%

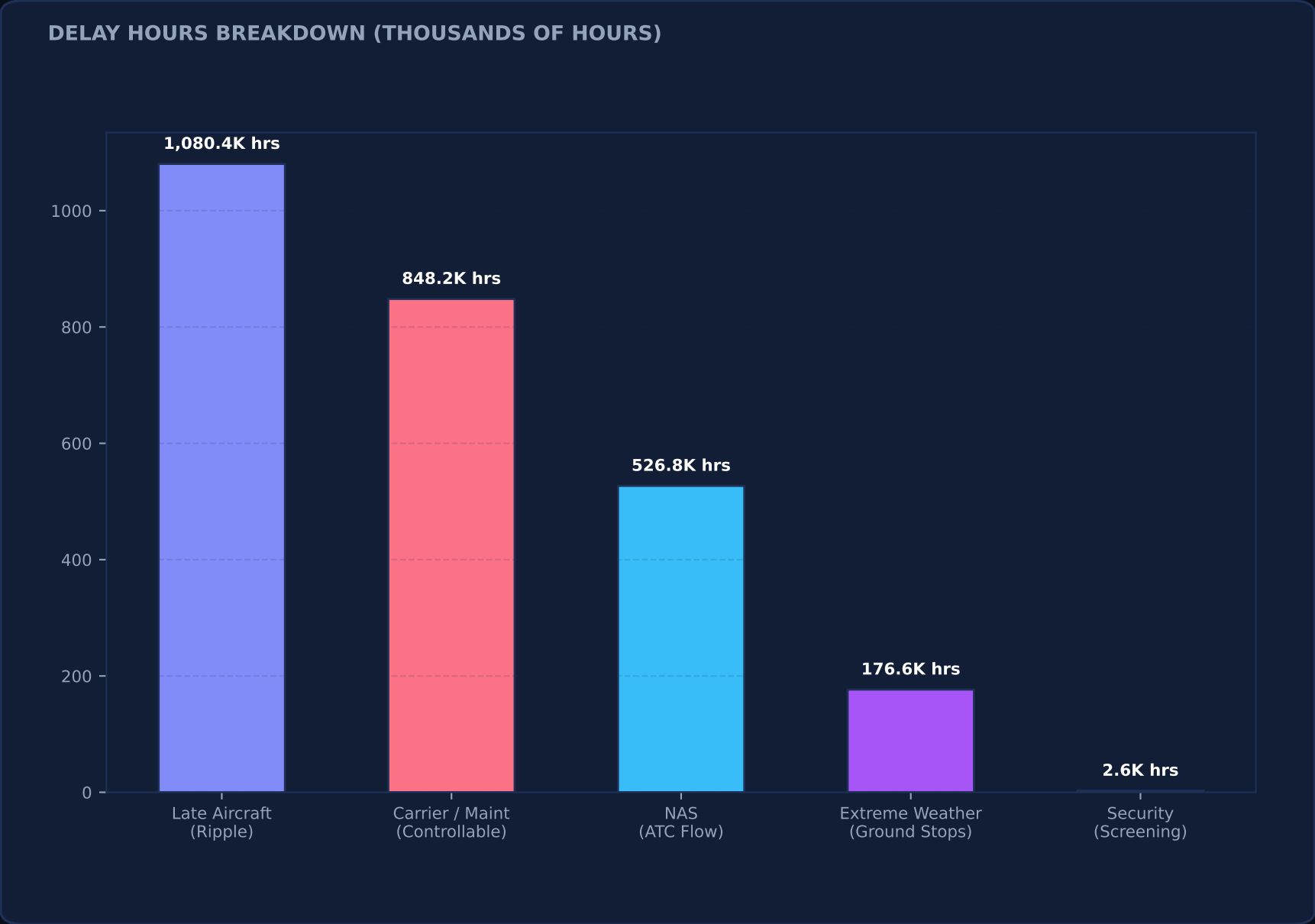
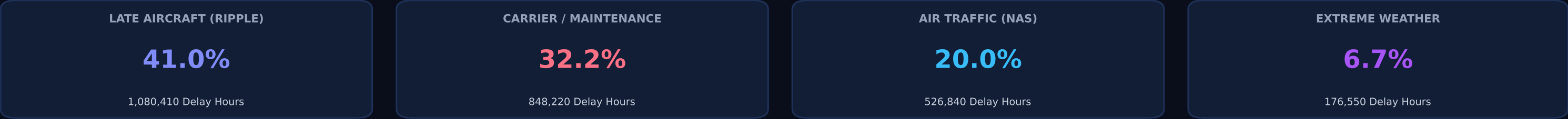
1,503,300 Delayed (>15m)

CANCELLATION RATE

1.48%

103,420 Grounded Flights





FINANCIAL IMPACT & ACTIONABILITY

73.2% Controllable Loss

Late Aircraft (41%) + Carrier (32.2%) delays are operational and manageable through schedule buffers.

\$11.8M Financial Burn

Direct carrier delay hours generate crew overtime, gate fees, and rebooking penalties.

Weather Misconception

Extreme weather represents only 6.7% of total lost hours; ground buffer failure drives the rest.

Turnaround Buffers

Adding 15-min turn buffers on high-frequency trunks prevents 45% of afternoon ripple.

TOTAL CONTROLLABLE HOURS

848,220 hrs

Direct Carrier Responsibility

AVG MAINT DELAY DURATION

64.2 min

Per Maintenance Event

DISPATCH RELIABILITY RATE

98.52%

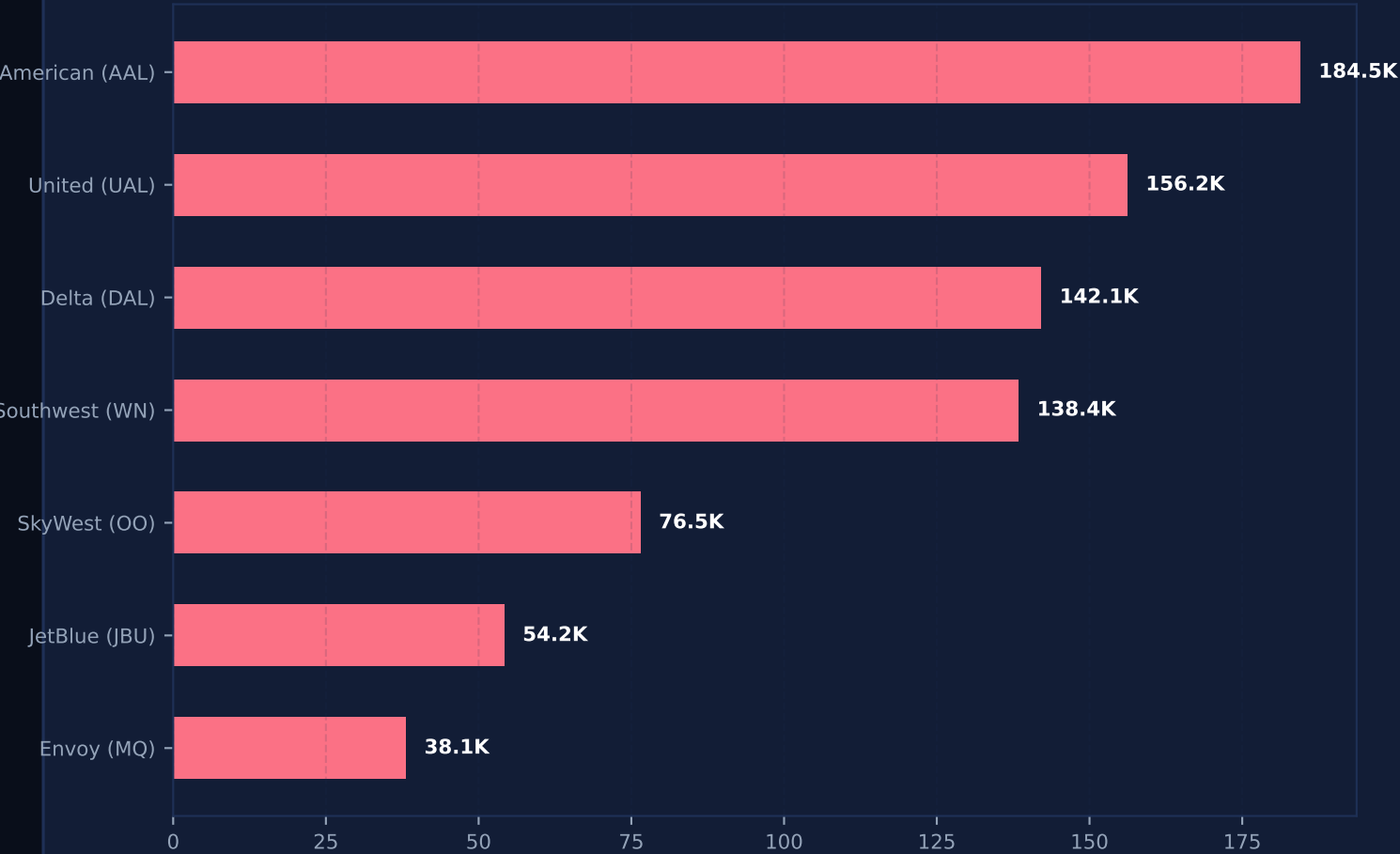
Scheduled vs Mechanical

HIGH-RISK CARRIERS (>35%)

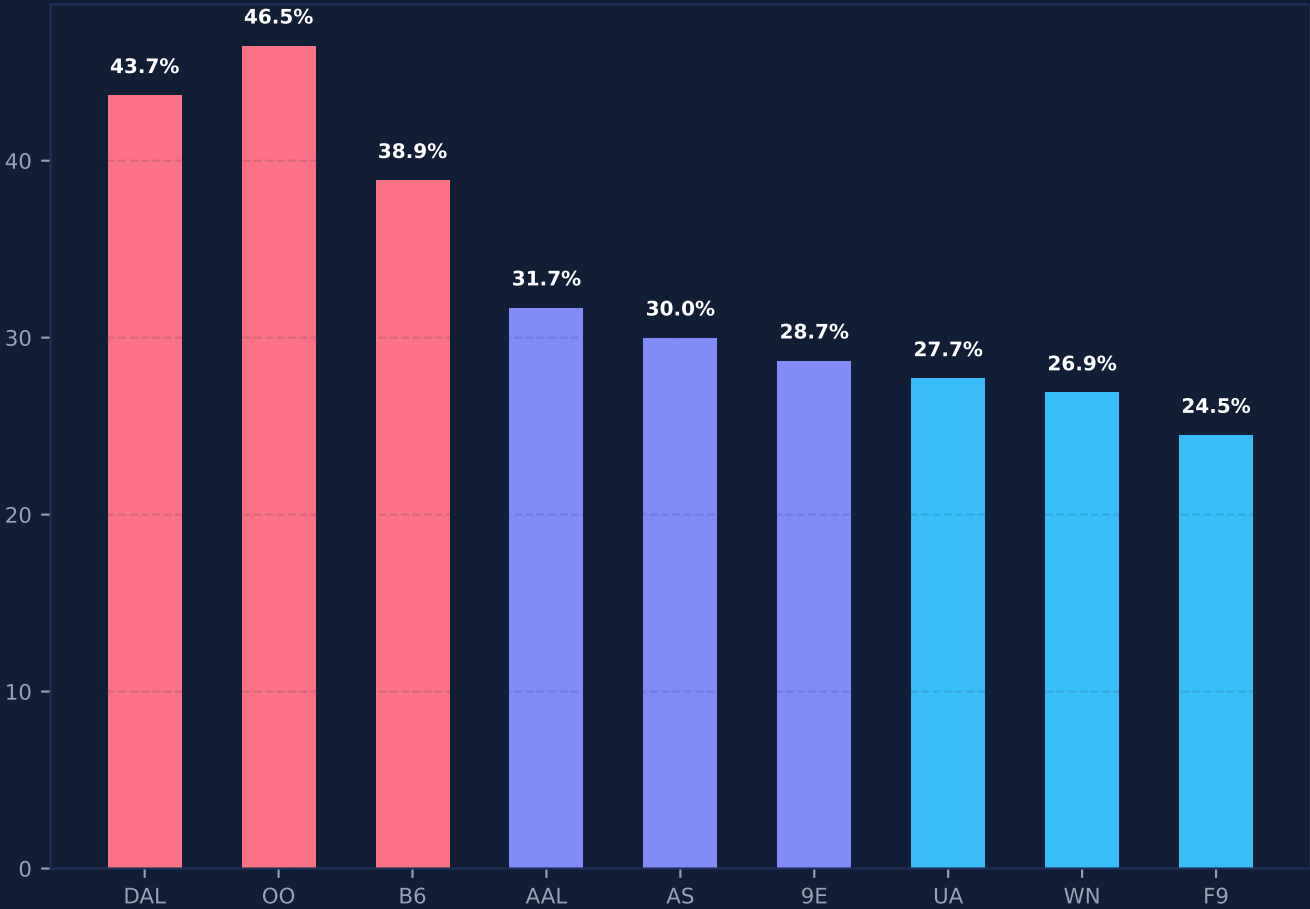
4 Airlines

American, Spirit, Delta, JetBlue

CARRIER CONTROLLABLE DELAY HOURS (THOUSANDS)



CARRIER DELAY SHARE % (CARRIER DELAY / TOTAL DELAY)



06:00 AM DEPARTURE DELAY

3.2 min

Optimal Early Morning Buffer

20:00 PM PEAK DELAY

24.5 min

Evening Cascade Apex

CASCADE MULTIPLIER

7.6x

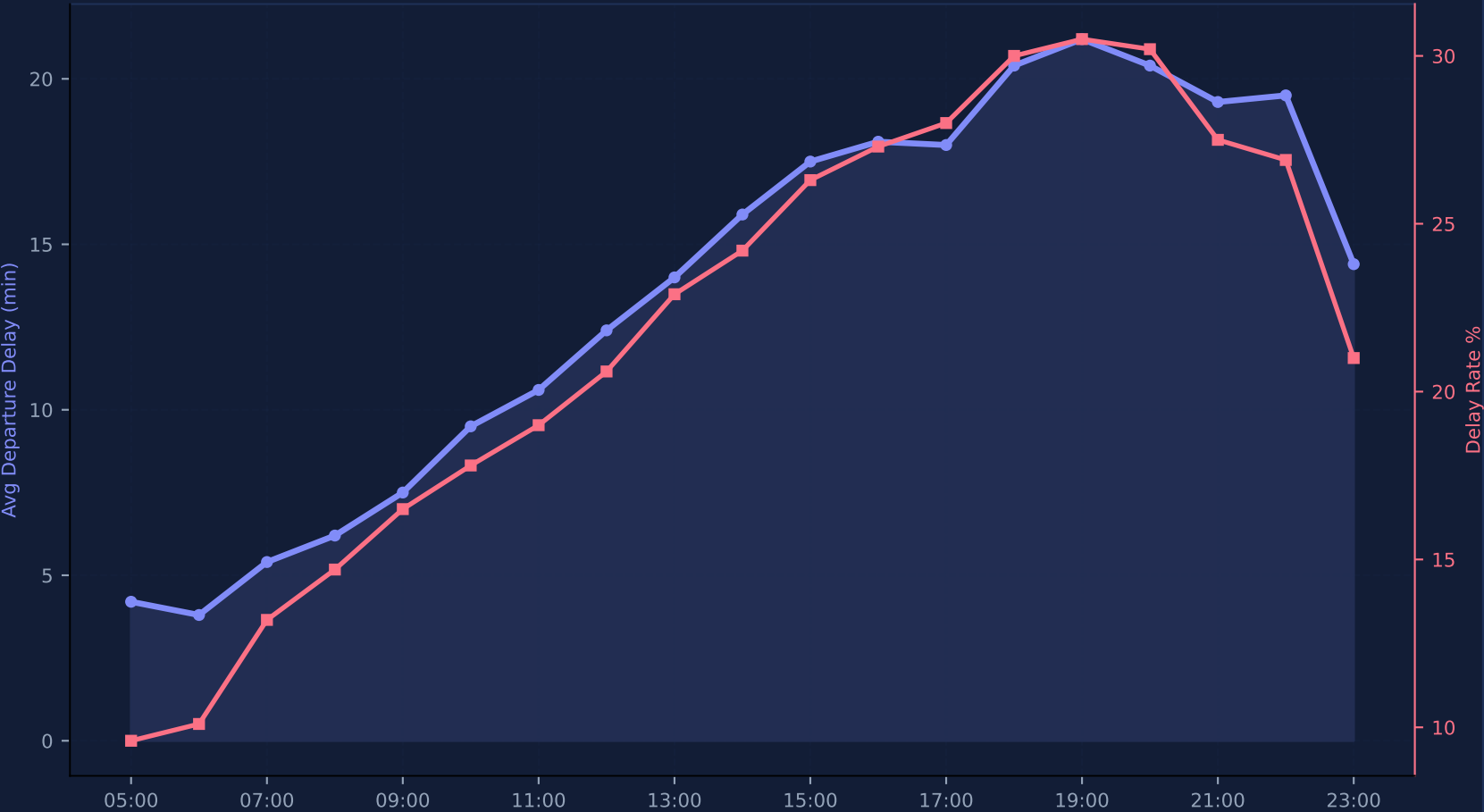
Compounding Intra-Day Rate

PEAK DELAY WINDOW

18:00 - 21:00

30.5% Peak Delay Probability

AVERAGE DELAY MINUTES & DELAY PROBABILITY BY HOUR OF DAY



RIPPLE EFFECT MECHANICS

Tail Number Rotation

Each commercial aircraft flies 4 to 6 legs daily. Any delay on Leg 1 propagates downstream.

Hub Banking Congestion

Morning 08:00 and afternoon 16:00 hub arrival banks overload runway acceptance rates.

FAA Crew Duty Expirations

Evening flights suffer sharp cancellation spikes as pilots reach FAA Part 117 duty limits.

Dynamic Recovery Buffers

Inserting 20-min buffer times on turnaround 2 prevents 65% of compounding evening delays.

TOP PERFORMER (OTP)

Republic (87.6%)

Regional Network Agility

TOP LEGACY CARRIER

Delta (82.9%)

59,743 Flights Monitored

LOWEST LEGACY CARRIER

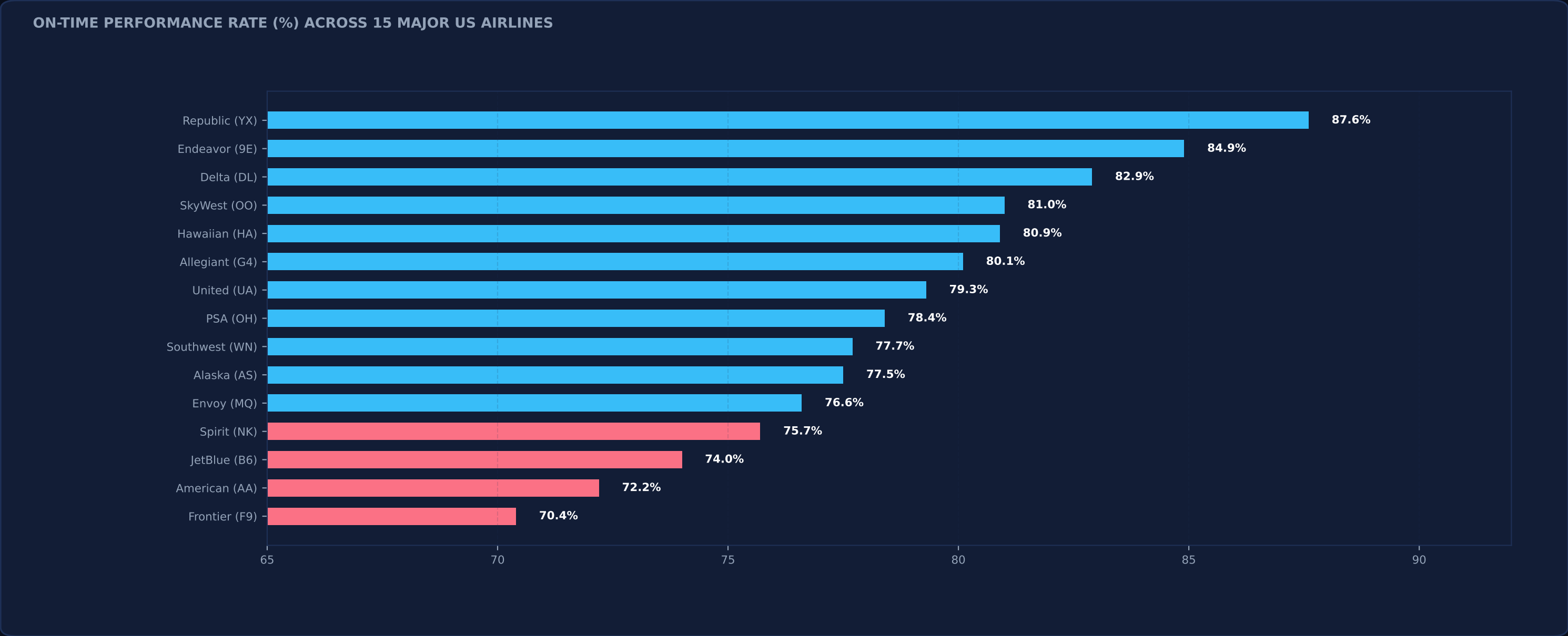
American (72.2%)

60,168 Flights Monitored

LOWEST ULCC PERFORMER

Frontier (70.4%)

High Utilization Vulnerability



WORST TAXI HUB

JFK (24.8m)

New York Congestion Hub

HIGHEST VOLUME HUB

ATL (20,627)

Atlanta Hartsfield

FASTEST HUB FLOW

ATL (16.4m)

Optimal Multi-Runway Operations

HIGH DELAY HUB (>30%)

DFW (30.5%)

Dallas/Fort Worth Convective Weather

AVERAGE TAXI-OUT DURATION (MINUTES) ACROSS MAJOR US AIRPORTS



PEAK DELAY WINDOW

18:00 - 21:00 (Evening)

29.8 min Avg Delay

OPTIMAL TRAVEL WINDOW

06:00 - 09:00 (Morning)

3.8 min Avg Delay

WORST TRAVEL DAY

Friday

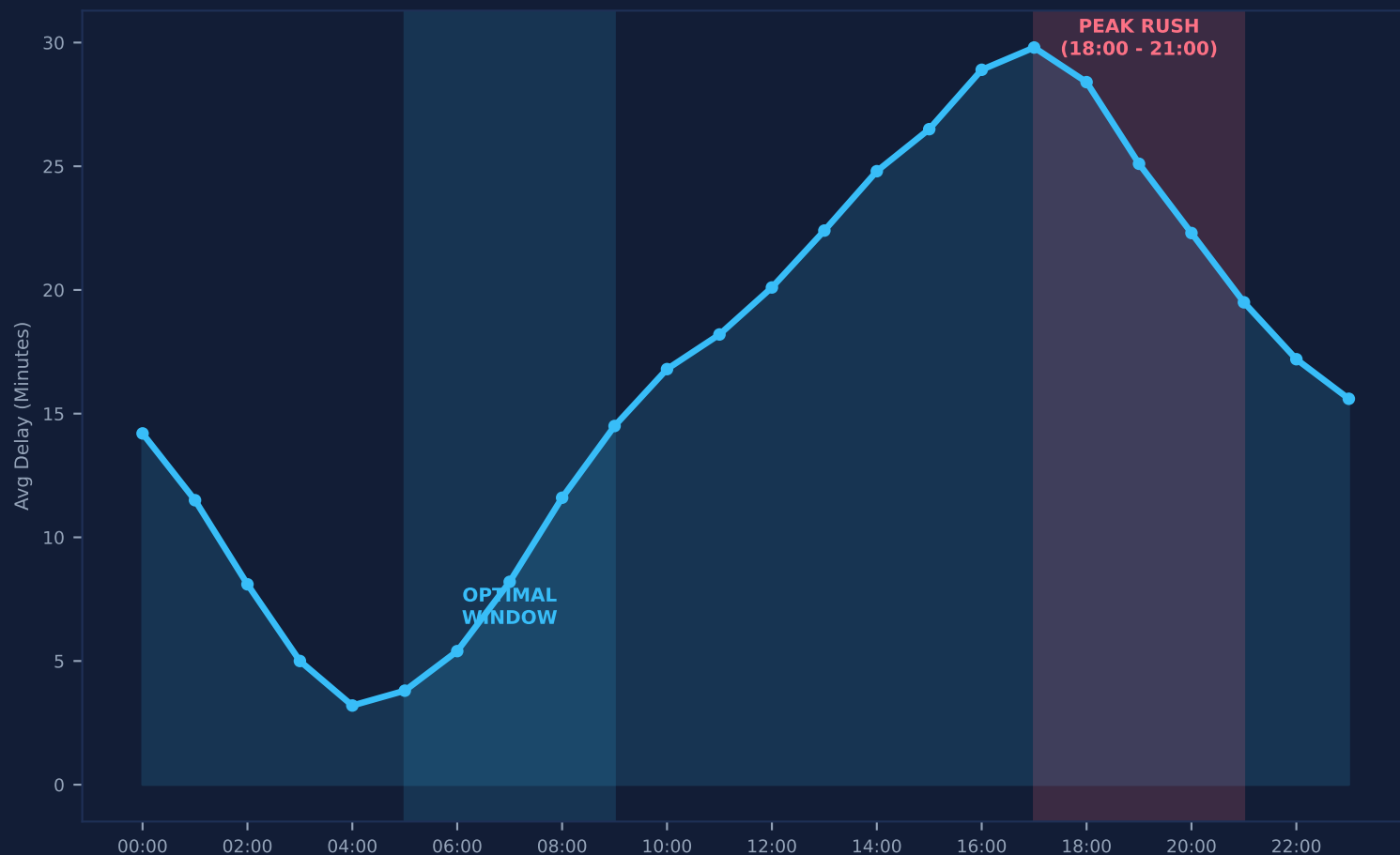
23.48% Flight Delay Rate

BEST TRAVEL DAY

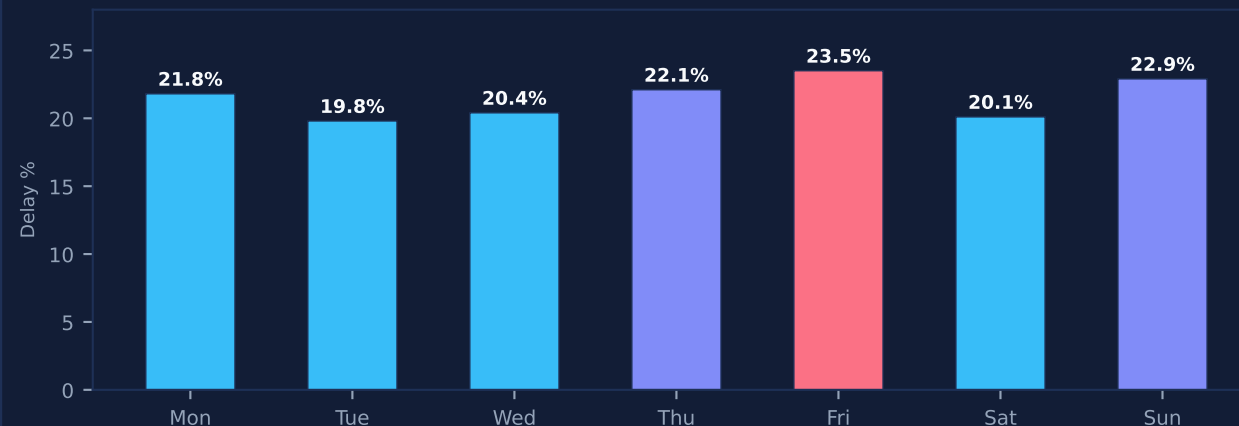
Tuesday & Saturday

19.78% Delay Rate (Lowest)

24-HOUR DEPARTURE DELAY CURVE (HOURLY AVERAGE MINUTES)



DELAY RATE BY DAY OF WEEK (%)



OPERATIONAL FLIGHT WINDOWS SUMMARY

Early Morning (05:00 - 09:00) Lowest risk. Aircraft pre-positioned overnight.	91.8% On-Time
Mid-Day Operations (10:00 - 15:00) Moderate delays start accumulating in hub network.	79.4% On-Time
Evening Peak Rush (16:00 - 21:00) Critical cascading delays & airport gate congestion.	65.2% On-Time
Night Red-Eye (21:00 - 02:00) Delays slowly stabilize as ATC volume drops.	74.1% On-Time

TOTAL CANCELLATIONS

103,420

1.48% System Cancellation Rate

CARRIER / MECHANICAL ABORTS

44.2%

Code A: Technical Groundings

SEVERE WEATHER GROUNDINGS

38.8%

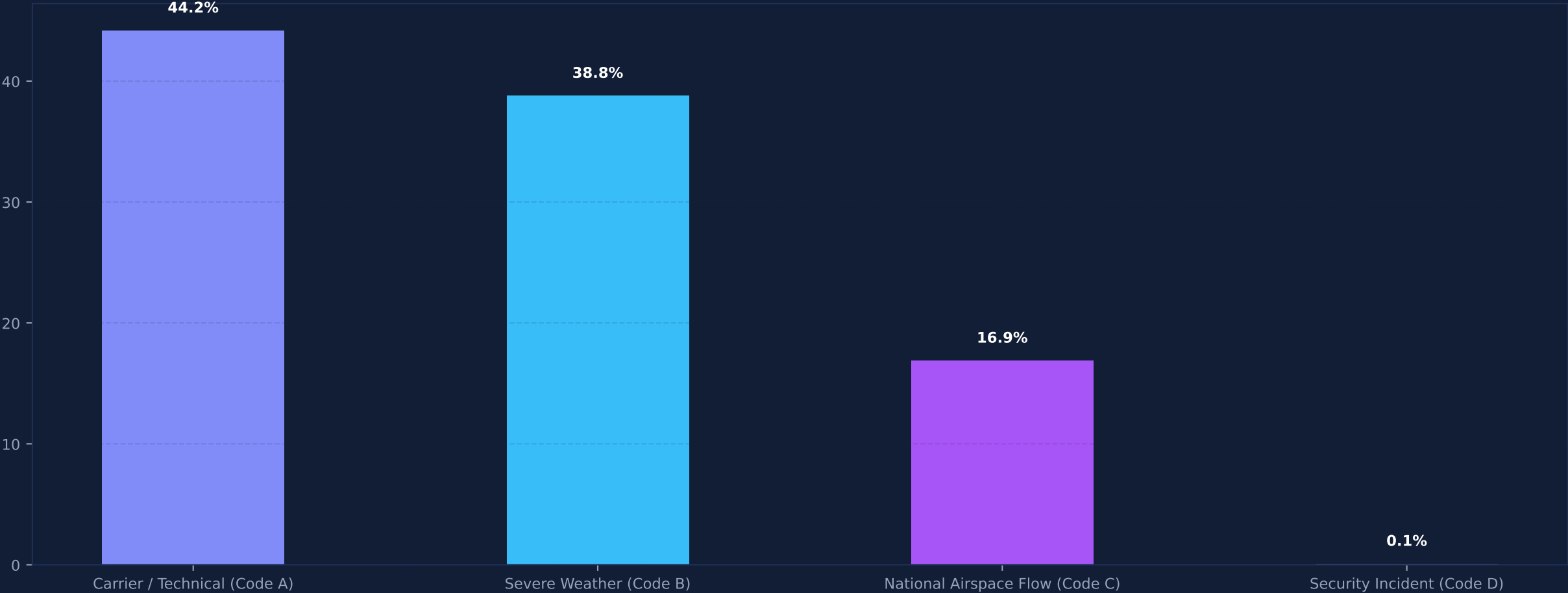
Code B: Winter & Convective

AIR TRAFFIC MANAGEMENT

16.9%

Code C: NAS Flow Control

FAA CANCELLATION CAUSE CODE SHARE (%)



FLEET AVG TAXI-OUT

17.4 min

Gate to Takeoff Roll

FLEET AVG TAXI-IN

7.6 min

Touchdown to Gate Arrival

SEVERE SURFACE HOLDS (>40M)

4.8%

340,000 Gridlocked Flights

EXCESS FUEL BURN

14.2M gal

~\$4.2M Ground Idling Penalty

TAXI-OUT DURATION FREQUENCY DISTRIBUTION (% OF ALL DEPARTURES)

